

FLEET RULE OF THE ROAD TEST JUN 14

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script.

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90%.

Answers are provided for uniformity of marking. This test is completely scenario based.

You are the OOW in HMS SOMERSET deployed on OP KIPION in the Gulf of Oman. You have just sailed from Fujairah, it is the morning watch with clear skies and good visibility but METO did forecast haze later in the day. As you pass through the breakwaters you slow down to disembark the pilot and a dhow which departed shortly after you begins to close.

1. The port lookout reports the dhow is dragging a net through the water each side of her. How would you classify this dhow (explain)? (R.3) (1)

2. What shapes would you expect the dhow to be displaying? (R.26) (1)

3. The dhow is closing on a steady bearing from R150. Who is the give way vessel and why? (R.13) (2)

The dhow passes clear just as the pilot boat pulls up alongside. As you make your departure from the port you are actually navigating on the left side of the channel because of local routing rules.

4. Who can make such local rules and what caveat is put on them? (R.1) (2)

You pass the extremity of port limits and the NO hands the ship over to you. He shows you were the RV for your replenishment this afternoon is on WECDIS and tells you he wants you to get there quickly.

5. What factors would you (and all vessels) take into account when deciding on a safe speed? (R.6)
(6)

The lookout reports to you that there is a vessel at R5 on the horizon heading towards you.

6. (a) What factors would you consider when ascertaining whether a risk of collision exists? (R.7) **(2)**

- (b) If there is any doubt what should you do? (R.7) **(1)**

- (c) What should you not make assumptions based upon? (R.7) **(1)**

You look at the vessel through binoculars and can see his foremast and mainmast almost in a line.

7. (a) What kind of situation is developing? (R.14) **(1)**

- (b) What action will you take? (R.14 & R.34a) **(2)**

8. (a) What should be the result of your action in Q.7? (R.8) **(1)**

- (b) How will you assess if your action in Q.7 is sufficient? (R.8) **(1)**

You pass the other vessel at a safe distance just as the haze the METO forecast earlier makes an unwelcome appearance reducing visibility to around 1Nm.

9. What condition do you now find yourself navigating in? (R.3) **(1)**

10. What speed and what machinery configuration should you now adopt? (R.19)
(2)

You have a radar contact at R30 range 6 miles that appears to be tracking from left to right close ahead of you.

11. What action do you take, when do you take it and what specifically should be avoided? (R.19)
(3)

Interrogation of the AIS data in WECDIS shows that this vessel is a coastal tanker of 112m length and her status is simply set to *Underway using Engine*.

12. At what range would you expect to hear her restricted visibility sound signal? (Annex III)
(1)

13. What sound signal would you expect to hear from her? (R.35) (1)

Before you have a chance to take any action, the contact gets to a range of 3 miles, still at R30 and with a relative vector passing close ahead of you, the haze lifts enough for you to see her and it is in fact a coastal tanker underway and making way.

14. Does the fact the visibility has lifted sufficiently for you to see her change the action you would take, what would you do differently and why? (R.15) (2)

The other vessel passes clear astern and quick as it arrived, the haze lifts completely and the visibility is up to around 10Nm. In the improved visibility, the starboard lookout reports a number of contacts spread out from G10 to G40. He thinks some of the closer ones are displaying what looks like two triangles pointing towards each other.

15. What are these vessels likely to be? (R.26) (1)

None of the reported vessels have a CPA of less than 0.5Nm and so after briefing the CO you maintain your course and speed. As you pass, two of the vessels are very close together each flying Flag Tango.

16. What specifically might these vessels be engaged in? (ICS) (1)

You pass all of the vessels safely and hand over the watch to head below for lunch. Later, you're back on watch for the evening, the sun has long since set and with no moon it is a particularly black night and good for star spotting. You see the following lights at G45:

W R
 R
 W

17. (a) What type of vessel is it? (R.30) (1)

- (b) What shape(s) would this vessel display by day? (R.30) (1)

You pass the vessel without incident when you sight the following lights at R30:

 W
 W
 W

 W W G W

18. (a) What type of vessel is it? (R.24) (3)

- (b) What shapes would you see by day? (R.24) (3)

You establish this vessel is on a steady bearing with a decreasing range.

19. Who is the give way vessel? (R.15) (1)

The range has decreased to only 2 miles, the bearing remains steady and neither vessel has acted.

20. (a) What may you do and when may you do it? (R.17) (1)

(b) If you took this action, what should you avoid? (R.17) (1)

The range has decreased further to just under 1Nm without either vessel taking any action. Risk of collision is rapidly becoming a very real risk.

21. What action shall you take? (R.17) (1)

You pass the vessel at a safe distance and the lookout reports the following lights fine to port on the horizon:

W
R
R
W
G R

22. (a) What type of vessel is it? (R.27) (1)

(b) What shapes would this vessel display by day? (R.27) (1)

You ascertain the bearing is steady and the range to be approximately 6 miles. You also determine that you are the give-way vessel under R.18.

23. What do the Rules dictate your action should be and when should you take it? (R.16) (1)

24. Are you justified in turning to port to avoid this vessel? Why? (R.3,14,16&18) (2)

You pass the other vessel at a safe distance and hand the watch over to head below for a well earned rest and contemplate the busy day you’ve just experienced.

You are the OOW in HMS SOMERSET deployed on OP KIPION in the Gulf of Oman. You have just sailed from Fujairah, it is the morning watch with clear skies and good visibility but METO did forecast haze later in the day. As you pass through the breakwaters you slow down to disembark the pilot and a dhow which departed shortly after you begins to close.

1. The port lookout reports the dhow is dragging a net through the water each side of her. How would you classify this dhow (explain)? (R.3) (1)

A vessel engaged in fishing (1) because:

(R.3(d)) The term vessel engaged in fishing means any fishing with nets, lines, trawls or other fishing apparatus which restrict manoeuvrability, but does not include a vessel fishing with trolling lines or other fishing apparatus which does not restrict manoeuvrability.

Stating it is a trawler is acceptable although the word trawler is only mentioned in R.26 and not defined in R.3 like other types of vessel are.

2. What shapes would you expect the dhow to be displaying? (R.26) (1)

(R.26(c)) ...a shape consisting of two cones with apexes together in a vertical line one above the other (1).

3. The dhow is closing on a steady bearing from R150. Who is the give way vessel and why? (R.13) (2)

The dhow is (1) because:

(R.3(a)) Notwithstanding anything contained in the Rules of part B, sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken (1).

The dhow passes clear just as the pilot boat pulls up alongside. As you make your departure from the port you are actually navigating on the left side of the channel because of local routing rules.

4. Who can make such local rules and what caveat is put on them? (R.1) (2)

(R.1(b)) Nothing in these Rules shall interfere with the operation of special rules made by an appropriate authority (1) for roadsteads, harbours, rivers, lakes or inland waterways connected with the high seas and navigable by seagoing vessels. Such special rules shall conform as closely as possible to these Rules (1).

You pass the extremity of port limits and the NO hands the ship over to you. He shows you were the RV for your replenishment this afternoon is on WECDIS and tells you he wants you to get there quickly.

5. What factors would you (and all vessels) take into account when deciding on a safe speed? (R.6) (6)

(R.6(a)) By all vessels:

(i) the state of visibility (1);

(ii) the traffic density including concentrations of fishing vessels or any other vessels (1);

(iii) the manoeuvrability of the vessel with special reference to stopping distance and turning ability in the prevailing conditions (1);

(iv) at night the presence of background light such as from shore lights or from backscatter of her own lights (1);

(v) the state of wind, sea and current, and the proximity of navigational hazards (1);

(vi) the draught in relation to the available depth of water (1).

The lookout reports to you that there is a vessel at R5 on the horizon heading towards you.

6. (a) What factors would you consider when ascertaining whether a risk of collision exists? (R.7) (2)

(R.7(d)) In determining if risk of collision exists the following considerations shall be among those taken into account:

- (i) **such risk shall be deemed to exist if the compass bearing of an approaching vessel does not appreciably change (1);**
- (ii) **such risk may sometimes exist even when an appreciable bearing change is evident, particularly when approaching a very large vessel or a tow or when approaching a vessel at close range (1).**

(b) If there is any doubt what should you do? (R.7) (1)

(R.7(a)) ...If there is any doubt such risk shall be deemed to exist (1)

(c) What should you not make assumptions based upon? (R.7) (1)

(R.7(c)) Assumptions shall not be made on the basis of scanty information (½), especially scanty radar information (½).

You look at the vessel through binoculars and can see his foremast and mainmast almost in a line.

7. (a) What kind of situation is developing? (R.14) (1)

(R.14(b)) Head-on situation (1)

(b) What action will you take? (R.14 & R.34a) (2)

(R.34(a)) Sound one short blast (1)

(R.14(a)) Alter course to starboard (1) to pass on the port side of the other

8. (a) What should be the result of your action in Q.7? (R.8) (1)

(R.8(d)) Action to avoid collision with another vessel shall be such as to result in passing at a safe distance (1)

(b) How will you assess if your action in Q.7 is sufficient? (R.8) (1)

(R.8(d)) ...The effectiveness of the action shall be carefully checked (½) until the other vessel is finally past and clear (½).

You pass the other vessel at a safe distance just as the haze the METO forecast earlier makes an unwelcome appearance reducing visibility to around 1Nm.

9. What condition do you now find yourself navigating in? (R.3) (1)

Restricted visibility (1) because:

(R.3(l)) The term restricted visibility means any condition in which visibility is restricted by fog, mist, falling snow, heavy rainstorms, sandstorms or any other similar causes.

10. What speed and what machinery configuration should you now adopt? (R.19) (2)

(R.19(b)) Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility (1). A power-driven vessel shall have her engines ready for immediate manoeuvre (1).

You have a radar contact at R30 range 6 miles that appears to be tracking from left to right close ahead of you.

11. What action do you take, when do you take it and what specifically should be avoided? (R.19) (3)

**Determine if risk of collision exists (1),
Take avoiding action in ample time (1),
Avoid an alteration of course to port (1) because:**

(R.19d) A vessel which detects by radar alone the presence of another vessel shall determine if a close-quarters situation is developing and/or risk of collision exists . If so she shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible the following shall be avoided:

- (i) an alteration of course to port for a vessel forward of the beam, other than for a vessel being overtaken;**
- (ii) an alteration of course towards a vessel abeam or abaft the beam.**

Interrogation of the AIS data in WECDIS shows that this vessel is a coastal tanker of 112m length and her status is simply set to *Underway using Engine*.

- 12 At what range would you expect to hear her restricted visibility sound signal? (Annex III) (1)

(Annex III 1(c) table) 1.5Nm (1)

13. What sound signal would you expect to hear from her? (R.35) (1)

(R.35(a)) A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes (½) one prolonged blast (½).

Before you have a chance to take any action, the contact gets to a range of 3 miles, still at R30 and with a relative vector passing close ahead of you, the haze lifts enough for you to see her and it is in fact a coastal tanker underway and making way.

14. Does the fact the visibility has lifted sufficiently for you to see her change the action you would take, what would you do differently and why? (R.15) (2)

Yes (1). R.19 only applies to vessels not in sight of one another when navigating in or near an area of restricted visibility so this simply becomes a crossing situation. She is the give way vessel (R.15) (1) because you are on her starboard bow and you are to maintain your course and speed (R.17a) (1)

The other vessel passes clear astern and quick as it arrived, the haze lifts completely and the visibility is up to around 10Nm. In the improved visibility, the starboard lookout reports a number of contacts spread out from G10 to G40. He thinks some of the closer ones are displaying what looks like two triangles pointing towards each other.

15. What are these vessels likely to be? (R.26) (1)

Vessels engaged in fishing (or trawling) (1)

None of the reported vessels have a CPA of less than 0.5Nm and so after briefing the CO you maintain your course and speed. As you pass, two of the vessels are very close together each flying Flag Tango.

16. What specifically might these vessels be engaged in? (ICS) (1)

Pair trawling (1)

You pass all of the vessels safely and hand over the watch to head below for lunch. Later, you're back on watch for the evening, the sun has long since set and with no moon it is a particularly black night and good for star spotting. You see the following lights at G45:

W R
 R
 W

17. (a) What type of vessel is it? (R.30) (1)

A vessel aground (1)

- (b) What shape(s) would this vessel display by day? (R.30) (1)

Three balls in a vertical line (1)

You pass the vessel without incident when you sight the following lights at R30:

W
W
W

W W G W

18. (a) What type of vessel is it? (R.24) (3)

A power-driven vessel probably over 50m in length (1) engaged in a towing operation ($\frac{1}{2}$) where the length of tow is over 200m ($\frac{1}{2}$). It is towing an inconspicuous, partly submerged object ($\frac{1}{2}$) which is over 100m in length ($\frac{1}{2}$).

- (b) What shapes would you see by day? (R.24) (3)

The towing vessel would display a diamond (1) where it can best be seen and the inconspicuous, partly submerged object would display a diamond at or near the aftermost extremity of the object (1) and since the length of tow is over 200m, an additional diamond shape where it can best be seen and located as far forward as possible (1).

You establish this vessel is on a steady bearing with a decreasing range.

19. Who is the give way vessel? (R.15) (1)

She is (1) – she is not showing that she is RAM and therefore this is straight forward crossing situation.

The range has decreased to only 2 miles, the bearing remains steady and neither vessel has acted.

20. (a) What may you do and when may you do it? (R.17) (1)

(R.17(a)(ii)) The latter may, however, take action to avoid collision by her manoeuvre alone ($\frac{1}{2}$), as soon as it becomes apparent ($\frac{1}{2}$) to her that the vessel required to keep out of her way is not taking action in compliance with these Rules.

You could also consider sounding the signal prescribed in R.34(d) (5 short rapid blasts) to indicate you have failed to understand the other vessels intentions.

- (b) If you took this action, what should you avoid? (R.17) (1)

(R.17(c)) A power-driven vessel which takes action in a crossing situation in accordance with subparagraph (a)(ii) of this Rule to avoid collision with another power-driven vessel shall, if the circumstances of the case admit, not alter course to port (1) for a vessel on her own port side.

The range has decreased further to just under 1Nm without either vessel taking any action. Risk of collision is rapidly becoming a very real risk.

21. What action shall you take? (R.17) (1)

(R.17(b)) When, from any cause, the vessel required to keep her course and speed finds herself so close that collision cannot be avoided by the action of the give-way vessel alone ($\frac{1}{2}$), she shall take such action as will best aid to avoid collision ($\frac{1}{2}$).

You pass the vessel at a safe distance and the lookout reports the following lights fine to port on the horizon:

W

R
R

W
G R

22. (a) What type of vessel is it? (R.27) (1)

A vessel probably over 50m in length (½) not under command (½)

- (b) What shapes would this vessel display by day? (R.27) (1)

Two balls or similar shapes in a vertical line (1) where they can best be seen

You ascertain the bearing is steady and the range to be approximately 6 miles. You also determine that you are the give-way vessel under R.18.

23. What do the Rules dictate your action should be and when should you take it? (R.16) (1)

Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early (½) and substantial action (½) to keep well clear.

24. Are you justified in turning to port to avoid this vessel? Why? (R.3,14,16&18) (2)

Yes you are justified in turning to port (1). This is not a head-on situation as defined by R.14, this is responsibilities between vessel situation as defined by R.18 (1). The other vessel is showing the lights of a vessel NUC as defined by R.3f and therefore is not a PDV as defined by R.3b. R.14 deals specifically with two PDVs meeting on reciprocal or nearly reciprocal courses and mandates both vessels alter to starboard. A vessel NUC is unlikely to be able to achieve this and so the onus is firmly on you to avoid a close-quarters situation or a collision.

You pass the other vessel at a safe distance and hand the watch over to head below for a well earned rest and contemplate the busy day you've just experienced.